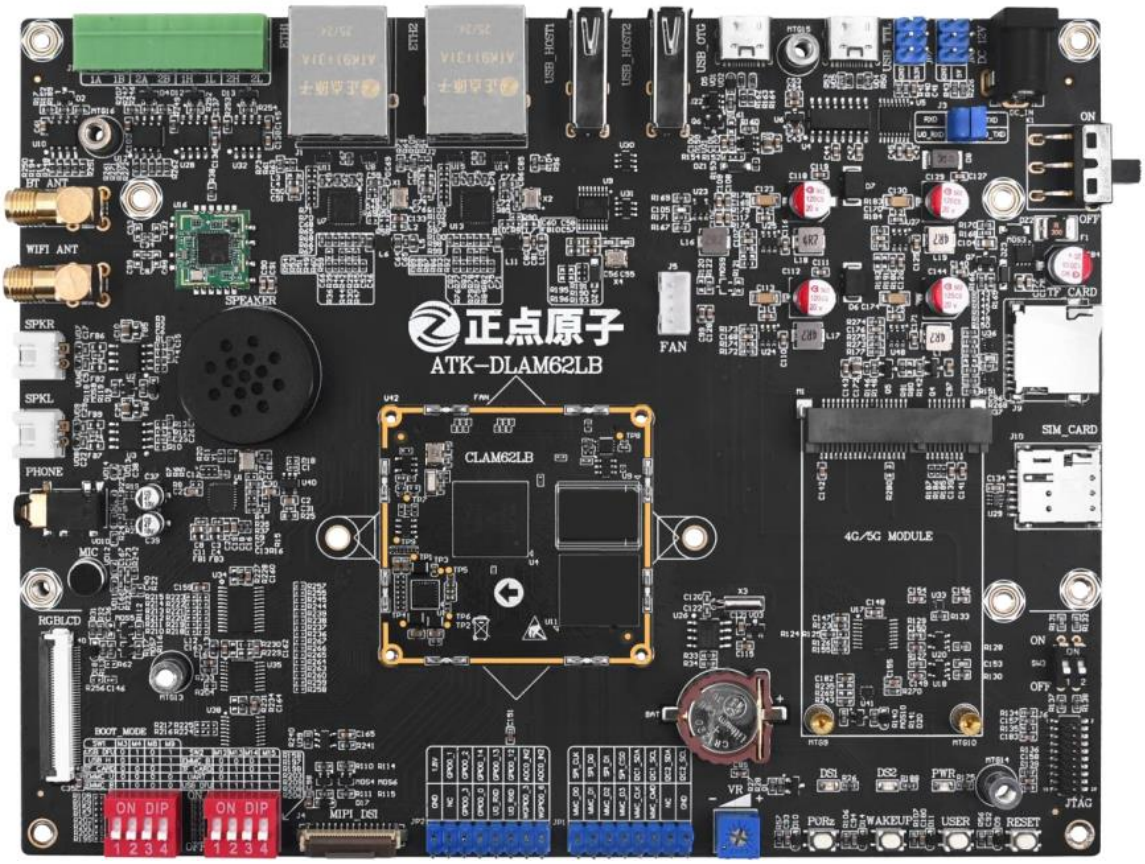
ATK-DLAM62LB

Development Board Specification

V1.0



ALIENTEK

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In order to get the latest version of product information, please regularly visit the download center or contact the customer service of Taobao ALIENTEK flagship store. Thank you for your tolerance and support.

Revision History:

Version	Version Update Notes	Responsible person	Proofreading	Date
V1.0	release officially	ALIENTEK Linux Team	ALIENTEK Linux Team	2026.03.01

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Chapter 1. Overview of Development Board

1.1 Introduction to the development board

The ATK-DLAM62B development board is a cost-effective board designed by ALIENTEK based on TI's AM62L chip (dual-core ARM Cortex-A53).

Consisting of a core board and a base board, the ATK-DLAM62LB development board adopts BTB connectors for interconnection. It is equipped with abundant peripheral resources and supports multiple interfaces, including I2C, SPI, CAN, NET, UART, MIPI DSI, RGB, McASP, USB2.0 and ADC.

It comes with comprehensive supporting software materials, including complete source code and detailed development documents. All software resources are available for download from ALIENTEK's cloud storage. To help enterprise customers improve development efficiency and shorten development cycles, ALIENTEK has compiled a full set of development resources for core board users, such as schematic diagrams, baseboard design files, mechanical dimensions, component footprints, connector specifications, factory system image source code, compilers and software packages, facilitating rapid product development for industrial applications.

Data Download Center: http://www.openedv.com/docs/boards/arm-linux/TI%20AM62L.html

Chapter 2. Development board hardware parameters

2.1 ATK-DLAM62LB development board

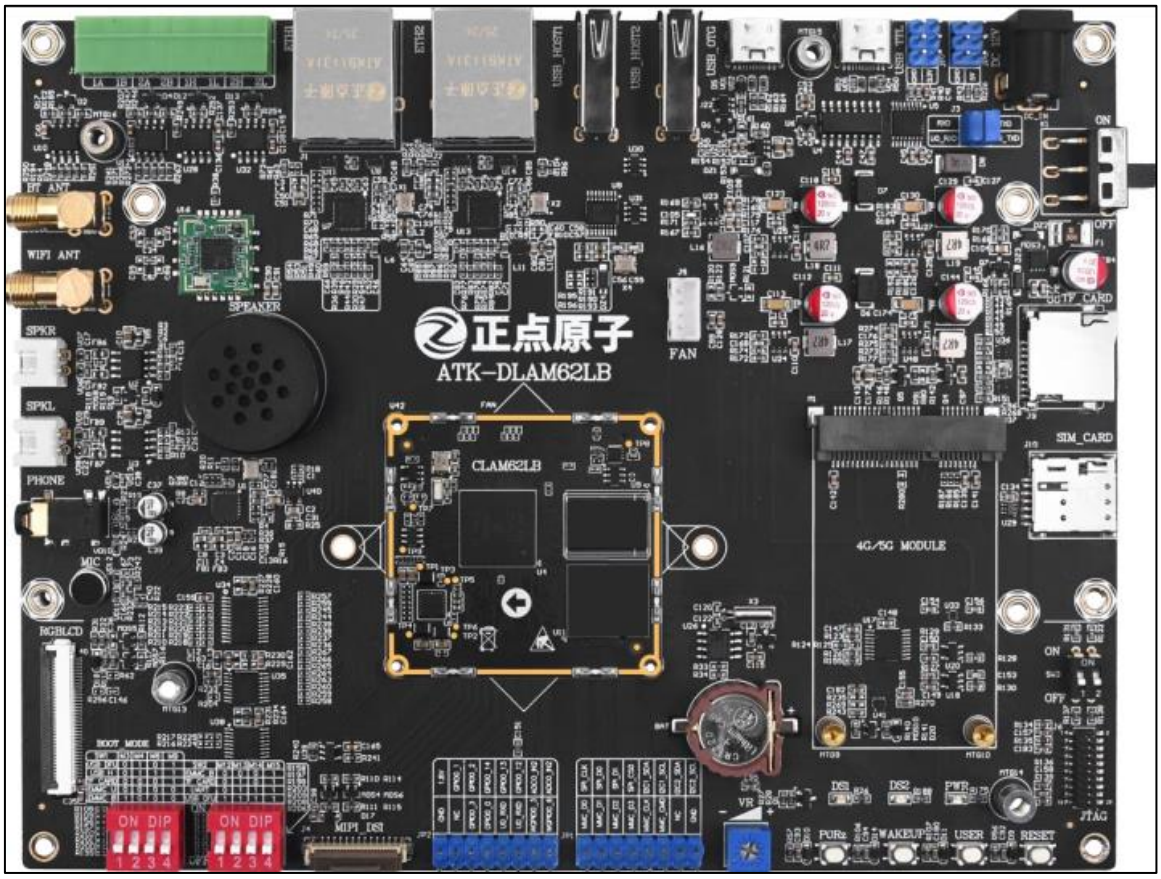

Figure 2.1-1 ATK-DLAM62LB development board

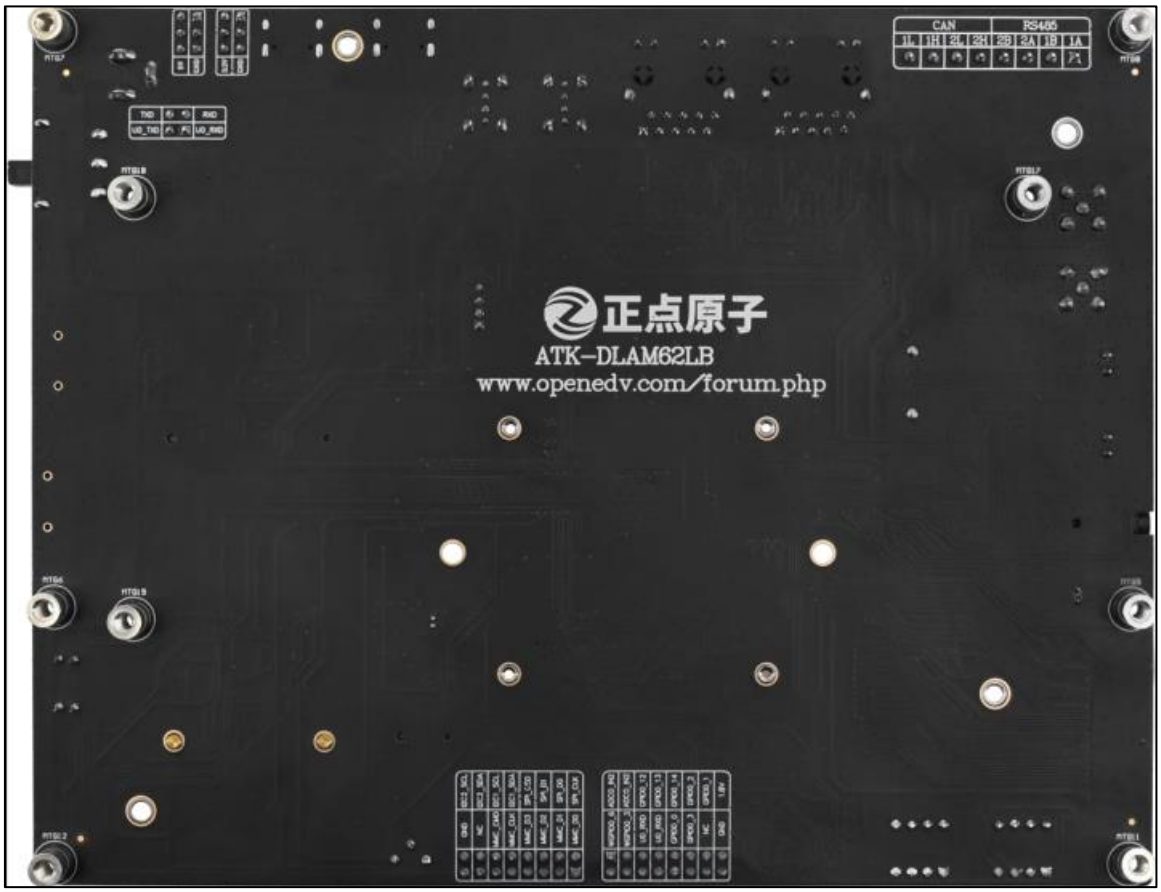

Figure 2.1-2 Back of the Development Board

2.2 ATK-DLAM62LB development board hardware parameters

Item	Parameter	Note
Dimensions and specifications	180mm*140mm	Length * width
Voltage of operation	12V	
Power consumption (1)	1.38W	Static power consumption, which depends on the peripherals
PCB process	4 layers, independent grounding signal layer	

Note:

(1) The power consumption data of the development board is measured with a 12V/1A input from the power supply, with only UART0 connected and no other peripherals attached. Actual power consumption depends on the peripherals connected to the development board.

2.3 ATK-DLAM62LB Development Board Onboard Resource Parameters

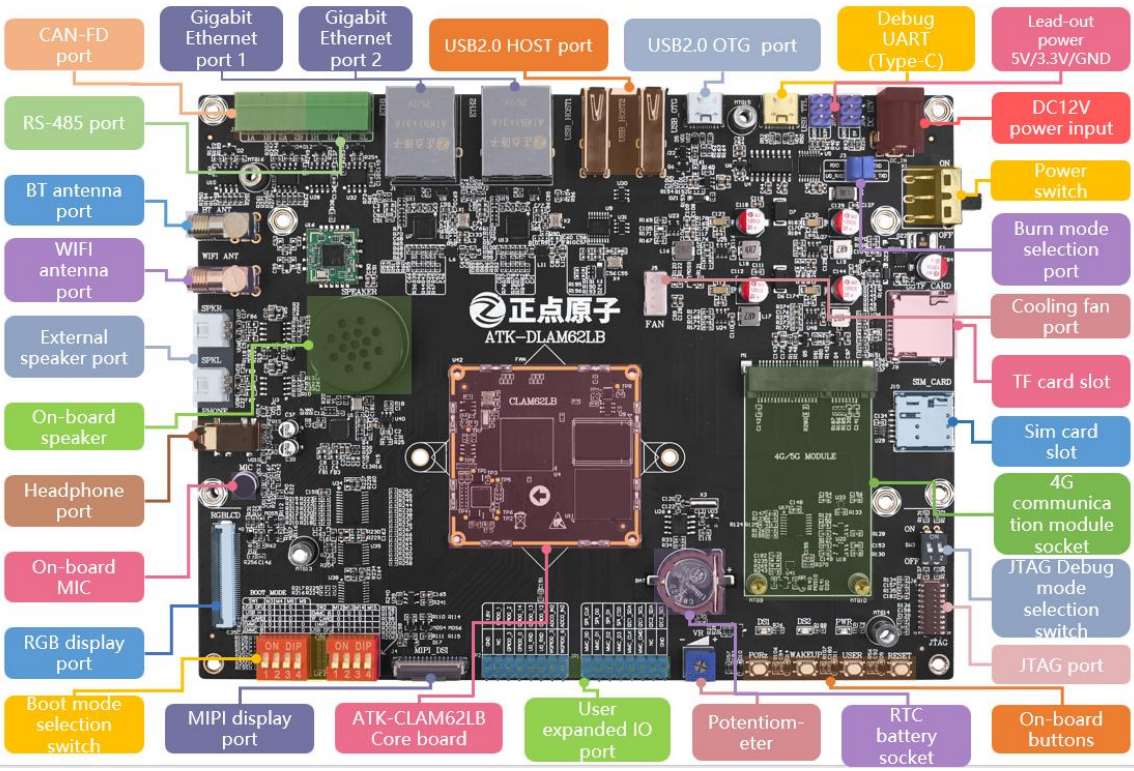

Figure 2.3-1 Development Board Resource Diagram

- 1 core board connector
- 1 DC power input connector (DC 12V)
- 1 power indicator LED (blue)
- 1 status indicator LED (red)
- 1 4G module indicator LED (green)
- 1 MIPI DSI interface
- 1 RGB interface
- 1 Mini PCIE 4G module interface
- 1 Nano SIM card interface
- 1 JTAG interface
- 2 reset buttons
- 2 user buttons
- 1 high-performance audio codec chip, ES8390
- 1 RTC chip
- 1 TF card interface
- 1 USB Wi-Fi & Bluetooth module, RTL8733BU
- 2 boot mode DIP switches
- 2 RS-485 interfaces
- 2 CAN interfaces
- 2 10M/100M/1000M Ethernet interfaces
- 2 USB HOST interfaces
- 1 USB OTG interface

- 1 USB-TTL interface
- 1 set of 5V power output interface
- 1 set of 3.3V power output interface
- 1 microphone (MIC)
- 1 small speaker
- 1 stereo audio headphone interface
- 2 external speaker interfaces for left and right channels
- 1 potentiometer
- 1 cooling fan interface

2.4 Factory System Default Configuration Signals

Factory System Default Configuration Signals: The factory system refers to the default out-of-box kernel firmware based on the development board that comes pre-installed on the core board. For detailed configurations, please refer to the device tree file of the factory system.

Table 2.4.1 Factory System Default Configuration Signals

Peripheral function	Quantity	Note
Ethernet port	2	Two Gigabit Ethernet ports are led out via RJ45 connectors.
Debug serial port	1	
McASP	1	Connects to the ES8390 audio chip.
RGB	1	Supports ALIENTEK 4.3/7.0/10.1-inch RGB screens.
MIPI DSI	1	Supports ALIENTEK 5.5/10.1-inch MIPI screens.
USB HOST	1	Expands two USB HOST interfaces.
USB OTG	1	
MMC	2	One channel is used for SD card, and the other is led out through pin headers.
RS-485	2	
CAN	2	
JTAG	1	
RTC	1	
4G	1	Supports GOSUNCN ME3630 / Quectel EC20
ePWM	3	
GPIO	Multi-channel	
ADC	1	

Chapter 3. Development board software resources

3.1 Factory system software resources

The factory Linux system software resources are shown in the following table:

Types	Description	Note
U-Boot	The version is 2025.01	Provide source code
Linux Kernel	The version is 6.12.24	Provide source code
Qt5	The version is 5.15.14	Provide source code
Cross compiler	aarch64-buildroot-linux-gnu-gcc	Used for compiling source code outside the SDK, such as drivers, applications, and QT source code.
System burn method	SD card burning	Provide tutorials
RGB LCD	RGB driver	Provide source code
LVDS	LVDS driver	Provide source code
MIPI DSI	MIPI DSI driver	Provide source code
Touch	GT9xx capacitive screen touch screen (only available at ALIENTEK)	Provide source code
Network	The Gigabit Ethernet PHY is YT8531	Provide source code
USB HOST	USB HOST 2.0	Provide source code
USB OTG	USB slave and host	Provide source code
4G module	GOSUNCN ME3630/ QUECTEL EC20	Provide source code
PMIC	TPS6521402	Provide source code
Function button	Two function buttons	Provide source code
RESET button	Reset function	Provide source code
External RTC	PCF8563 RTC chip	Provide source code
TF card /EMMC	SDMMC driver	Provide source code
LED	GPIO	Provide source code
Audio	ES8390	Provide source code
USB WIFI&BT	RTL8733BU	Provide source code
Serial port	USB debugging serial port, RS-485	Provide source code
ePWM	LCD PWM backlight	Provide source code
CAN	Support FD CAN	Provide source code
ADC	ADC Analog-to-Digital Conversion Interface	Provide source code

Chapter 4. Development documents

4.1 Documents

Development materials	
Document list (PDF format)	Description
ATK-DLAM62L_C Application Hardware Development Examples	Learning to operate development board hardware at the application layer
ATK-DLAM62L Embedded Qt Development Examples	Learning Qt examples based on the factory system
Fundamentals of Linux Embedded Qt Development	Learning the basics of Qt development based on the factory system
Linux_C Application Programming Reference Manual	Learning application programming based on the factory system
ATK-DLAM62L Factory System SDK Usage Guide	Learning to use the SDK
ATK-DLAM62L Embedded Linux Driver Development Reference Manual	Learning driver development based on the factory system

User manual	
Document list (PDF format)	Description
ATK-DLAM62L Development Board Quick Start Manual	1.System Flashing 2.Development Board Usage and Testing
ATK-DLAM62L Buildroot User Manual	Introduction to Buildroot File System Development
ATK-DLAM62L Development Board Hardware Reference Manual	1.Development Board Resource Description 2.Schematic Diagram Description
Ubuntu Environment Setup & VSCode Installation and Usage	Development Environment Construction
ADB Tool User Guide	ADB Installation and Usage
ATK-DLAM62L Qt Development Environment Setup	Installation and Configuration of Qt Creator and Related Environments
Development Board File Transfer Reference Manual	File Transfer Between Development Board and PC
ATK-DLAM62L Factory System UI Source Code Usage Guide	Tutorial for Using the Factory Source Code
ATK-DLAM62L Development Board GPIO User Guide	Introduction to GPIO Function Multiplexing
ATK-DLAM62L Development Board uEnv User Guide	Introduction to the uEnv File

ATK-DLAM62L Factory System Pin Multiplexing Modification Manual	Modification of Peripheral Functions for Development Board and Core Board Interfaces
Reference Manual for Configuring Static IP Based on Buildroot System	Static IP Configuration for the Development Board

Chapter 5. Development materials

Download development materials:

http://www.openedv.com/docs/boards/arm-linux/TI%20AM62L.html

Development materials are written based on ATK-DLAM62LB development board, please use the development board for project research and testing.

Development board Catalog:

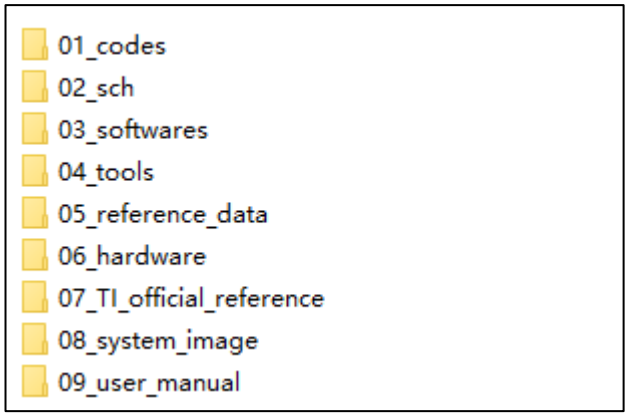

Figure 4.1-1 Development board catalog

Core board information Download:

The core board information is based on the ATK-CLAM62LB core board, extracted from the ATK-CLAM62LB development board information, convenient for users to download and use separately.

Core board data directory:

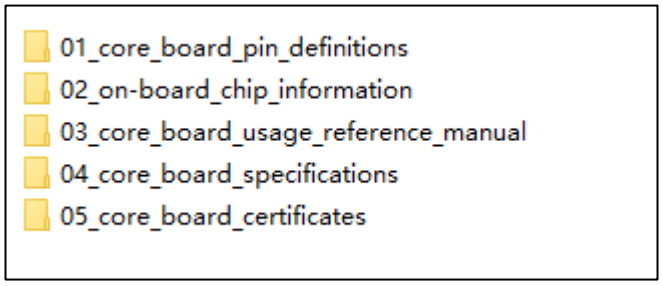

Figure 4.1-2 Core board data directory

Chapter 6. Development board structure size

6.1 Development Board Mechanical Drawing

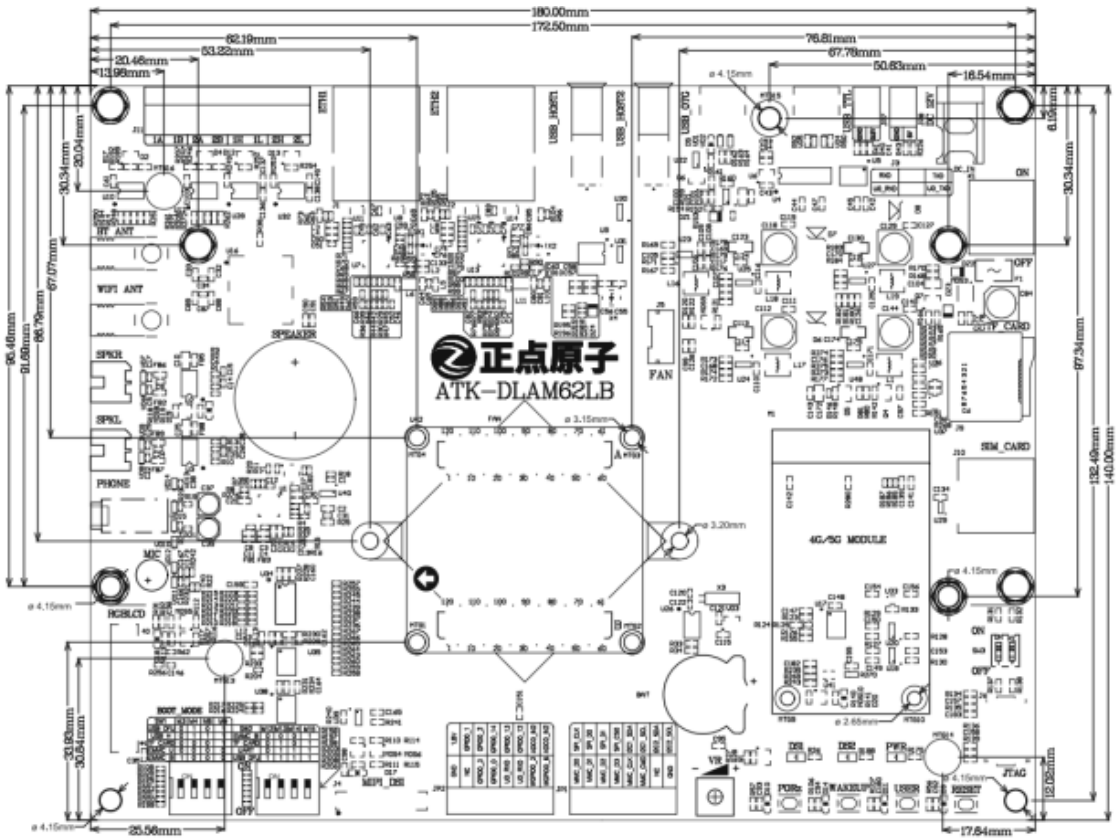

Figure 6.1-1 Development board structure size

Chapter 7. Optional accessories

7.1 ATK-MD0700-1024*600 screen

[RGB screen] ALIENTEK 7.0 inch RGB LCD module capacitive touch LCD display 1027*600

Purchase link:

https://detail.tmall.com/item.htm?abbucket=12&id=609758563397&rn=2bc286587344e22f9d93b3dfb05610f6&spm=a1z10.5-b-s.w4011-24686329149.89.1c3a48f5PH3kc9&skuId=4882572749742

Figure 7.1-1 ATK-MD0700-1027*600 screen

7.2 ATK-MD0550-7201280 screen

5.5 inch MIPI capacitive touch screen module 720*1280

Purchase link:

https://detail.tmall.com/item.htm?abbucket=12&id=691859000787&rn=0c60696d5f6381f7af67450da293e9fa&spm=a1z10.5-b-s.w4011-24686329149.129.3c74478c2YVEvx

Figure 7.2-1 ATK-MD0550-7201280 screen

Chapter 8. Precautions and maintenance

Notes

- Do not plug and unplug peripheral modules with power!
- Before using the product, please carefully read this manual and related development manuals, and pay attention to the applicable matters of the platform.
- Follow all instructions and warnings on the product.
- Please use this product in a cool, dry and clean place.
- Please keep the product dry. If any liquid splashes or soaks, power off immediately and let dry thoroughly.
- Do not use organic solvents or corrosive liquids to clean the product.
- Do not use or store this product in dusty, dirty and messy environment.
- If not used for a long time, please package this product, pay attention to moisture-proof and dust-proof.
- Pay attention to the ventilation and heat dissipation of the product during use to avoid component damage caused by excessive temperature during operation.
- Do not use this product in alternating hot and cold environment to avoid dew damage to components.
- Do not treat this product roughly, drop, knock or shake violently may damage the line and components.
- Pay attention to anti-static when using this product.
- FPC flexible cable is fragile, when plugging cable, pay attention to check whether the metal at both ends of the cable is misplaced and falling off.
- All products have passed the product test before shipment. Please use the development board corresponding to the ALIENTEK for power on test for the first time.
- Do not repair or disassemble the company's products by yourself. If the product fails, please contact the company in time for maintenance.
- Unauthorized modification or use of unauthorized parts may damage the product, the resulting damage will not be repaired.

Chapter 9. After sales service

9.1 Terms of after-sales service

1). After receiving the goods, please open them in front of the express, and sign after acceptance. If you find that the goods are less after signing, take photos in time and contact the seller's customer service to explain the situation within 15 days. If the feedback is lack of goods after 15 days, we will not reissue the goods. Other reasons notwithstanding).

2). 15 days -1 month: we are responsible for the return freight repair of product problems. Human factors damage expensive main chip or LCD screen, touch screen. The buyer needs to pay the cost and one time shipping fee, no maintenance fee.

3). 1-3 months: the problem of the product itself (non-human factors), we are responsible for the delivery of the past freight maintenance. If the main chip is burned out and the LCD screen and touch screen are damaged, the buyer needs to pay the cost, and the maintenance fee is not charged.

4) After 3 months: the buyer shall bear the return freight and the cost of chip, LCD screen and touch screen. No service charge.

9.2 After-sales Support

Technical support:

QQ group: ALIENTEK AM62x Communication group: 984075316

ALIENTEK AM62x User Group: 984080917 (order number required)

Taobao shop: ALIENTEK flagship store

Forum: http://www.openedv.com/thread-353774-1-1.html